

1. START

CLASS Person

DECLARE Name

DECLARE Age

END CLASS

CLASS Student INHERITS Person

Reg.No, Marks[5]

CALCULATE AVERAGE()

$$Avg = (M1 + M2 + M3 + M4 + M5) / 5$$

END

END CLASS

CLASS Research Student inherits Student

ResearchArea

DISPLAY DETAILS()

END

END CLASS

CREATE OBJECT R of Research Student

INPUT Name, Age, RegNo, 5 Marks, ResearchArea

CALL CALCULATE AVERAGE()

CALL DISPLAY DETAILS()

STOP

```
2. BEGIN
  READ N
  READ ARRAY A
  FOR I=0 TO N-1
    FOR J=I+1 TO N-1
      IF A[I]==A[J]
        A[J]=-1
      END IF
    END FOR
  END FOR
  PRINT Modified Array
END
```

END

3. CLASS Area  
METHOD area(length)  
RETURN length \* length  
END METHOD  
  
METHOD area(length, breadth)  
RETURN length \* breadth  
END METHOD  
END CLASS

4. BEGIN

sum  $\leftarrow$  0

FOR  $i \leftarrow 1$  TO 5

IF  $i \bmod 2 \neq 0$

sum = sum + i

ELSE

sum = sum - i

ENDIF

ENDFOR

PRINT SUM

o/p:  $i=1 \Rightarrow \text{sum} = 0 - 1 = -1$

$i=2 \Rightarrow \text{sum} = -1 + 2 = 1$

$i=3 \Rightarrow \text{sum} = 1 - 3 = -2$

$i=4 \Rightarrow \text{sum} = -2 + 4 = 2$

$i=5 \Rightarrow \text{sum} = 2 - 5 = -3$

-3 will be printed

5. BEGIN  
READ N  
READ Array A  
 $first \leftarrow -\infty$   
 $second \leftarrow -\infty$   
FOR  $i \leftarrow 0$  TO  $N-1$   
    IF  $A[i] > first$  THEN  
         $second \leftarrow first$   
         $first \leftarrow A[i]$   
    ELSE IF  $A[i] > second$  AND  $A[i] \neq first$  THEN  
         $second \leftarrow A[i]$   
    ENDIF  
END FOR  
  
PRINT second  
END

6.

BEGIN

INTERFACE payment

pay(amount)

END INTERFACE

CLASS Credit Card IMPLEMENTS PAYMENT

pay(amount)

PRINT

END CLASS

CLASS UPI IMPLEMENTS payment

Pay(amount)

PRINT

END CLASS

READ choice, amount

IF choice = 1

p ← Credit Card

ELSE IF choice = 2

p ← UPI

ELSE

p ← NetBanking

ENDIF

CALL p.pay(amount)

END

7. CLASS Animal  
METHOD speak()  
PRINT "Animal"

END METHOD

END CLASS

CLASS Dog inherits Animal

METHOD speak()

PRINT "Barks"

END METHOD

END CLASS

Animal obj ← new Dog()

obj.speak()

8. CLASS Employee

DATA

id

name

CONSTRUCTOR(id, name)

this.id  $\leftarrow$  id

this.name  $\leftarrow$  name

END CONSTRUCTOR

END CLASS

9. BEGIN

CLASS Vehicle

Vehicle No

owner Name

display()

END CLASS

CLASS Car Inherits Vehicle

FuelType

END CLASS

CLASS Truck inherits Vehicle

load Capacity

END CLASS

READ N

FOR I ← 1 TO N

READ Type

IF type = "Car" THEN

obj ← new Car

READ vehicleNo, ownerName, fuelType

ELSE

obj ← new Truck

READ vehicleNo, ownerName, load Capacity

ENDIF

STORE obj

ENDFOR

FOR EACH obj

CALL obj.display()

ENDFOR

END